

Identical Relation of Interpolation and Decimation in the Linear Canonical Transform Domain

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Abstract

The theorems of identical relation for interpolation and decimation in the Linear Canonical Transform (LCT) domain have been proposed in this paper. For improving the efficiency of sampling rate conversion and saving the memory in digital system, we propose two identical relations in sampling rate conversion, one in interpolation and the other one in decimation. The result shows that the two identical relations can properly secure the sampling rate conversion and reduce the operating works significantly. The simulation results are also presented to show the correctness of the proposed theorems. These two identical relations can be widely used in sampling rate conversion in the LCT domain.

Key words: identical relation, multirate sampling, linear canonical transform

1. Introductions

The theorem of multirate digital signal processing is put forward in how to convert sampling rate in digital systems. As an important branch of signal processing, multirate digital signal processing has made great progress both in fundamental theorem and practical application, which is widely applied in communication, voice processing, image compressing, antenna system, etc.

Traditional multirate signal processing is based on classical Fourier Transform. The Linear Canonical Transform (LCT), as the generalized form of the Fourier transform, can be applied in a wider range.

In practical application, we convert sampling rates by putting a filter after interpolation and a filter before decimation to avoid aliasing, as illustrated in the following figure. [1]

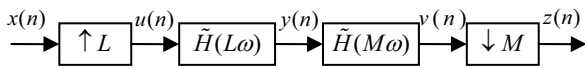


Fig.1 Traditional process in sampling rate conversion

However, this process will cause high operating works due to the complexity of the LCT. In order to reduce the operating works and improve the efficiency, we attempt to set the filter in the low sampling rate side, as in Fig.2. The validity of this transposition is proved in Fractional Fourier Transform domain [2], and it can reduce the operating works significantly. Whether it can be applied in LCT domain deserve further investigation.

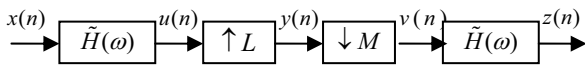


Fig.2 New process in sampling rate conversion

In this paper, we propose identical relation in interpolation and decimation in LCT domain, respectively, which can significantly reduce the operating works in sampling rates conversion. In section II, we will study the interpolation and decimation in LCT domain. In section III, we will propose the identical relations in interpolation and decimation. Meanwhile, these theorems will be proved. In section IV, simulation results of the given theorems will be presented. Finally in Section V, we give a conclusion to this paper.

2. Interpolation and decimation in LCT domain

2.1. The Linear Canonical Transform

The LCT of a continuous signal $x(t)$ is defined as [3]

$$X_{(a,b,c,d)}(u) = L_{(a,b,c,d)}(x(t))(u) = \begin{cases} \sqrt{\frac{1}{j2\pi b}} e^{j\frac{d}{2b}u^2} \int_{-\infty}^{\infty} x(t) e^{j(\frac{a}{2b}t^2 - \frac{1}{b}ut)} dt & b \neq 0 \\ \sqrt{d} e^{j\frac{cd}{2}u^2} x(du) & b = 0 \end{cases} \quad (1)$$

where $ad - bc = 1$ and we only consider the case $b \neq 0$ in this paper.

Consider the discrete time signal $x(n)$ sampled from the continuous signal $x(t)$ with the sampling period Δt , we can define the discrete time LCT (DTLCT) of $x(n)$ as [2]

$$\tilde{X}_{(a,b,c,d)}(u) = \sqrt{\frac{1}{j2\pi b}} e^{j\pi\frac{d}{2b}u^2} \sum_{n=-\infty}^{+\infty} x(n) e^{j(\frac{a}{2b}n^2\Delta t^2 - \frac{1}{b}un\Delta t)} \quad (2)$$

And the digital frequency in the LCT domain is defined as

$$\omega = \frac{1}{|b|} \cdot u \cdot \Delta t \quad (3)$$

we have the LCT of $x(n)$ denoted by ω :

$$\tilde{X}_{(a,b,c,d)}(\omega) = \sqrt{\frac{1}{j2\pi b}} e^{j\frac{d}{2b}(\frac{b\omega}{\Delta t})^2} \sum_{n=-\infty}^{+\infty} x(n) e^{j\frac{a}{2b}n^2\Delta t^2} e^{-j\pi\omega n \text{sgn}(b)} \quad (4)$$

Where

$$\text{sgn}(b) = \begin{cases} 1 & b > 0 \\ -1 & b < 0 \end{cases}$$

The corresponding inverse transform is

$$x(n) = \sqrt{\frac{jb}{2\pi}} e^{-j\frac{d}{2b}\pi^2\Delta t^2} \int_{-\pi}^{\pi} \tilde{X}_{(a,b,c,d)}(\omega) e^{-j\frac{d}{2b}\left(\frac{b\omega}{\Delta t}\right)^2} e^{j\pi\omega\text{sgn}(b)} d\omega \quad (5)$$

2.2. Interpolation and Decimation

Interpolation and decimation are two basic operations in multirate digital signal processing. In this subsection, we will introduce interpolation and decimation in LCT domain.

To increase the sampling rate of the origin signal $x(n)$ by an integer factor L , the output sequence $y(n)$ is

$$y(n) = \begin{cases} x(\frac{n}{L}), n = Lk, k \in Z \\ 0, n \neq Lk, k \in Z \end{cases} \quad (6)$$

The sampling period of $y(n)$ is reduced to $\Delta t_y = \Delta t_x / L$, and the DTLCT of $y(n)$ is

$$\tilde{Y}_{(a,b,c,d)}(\omega) = \tilde{X}_{(a,b,c,d)}(L \cdot \omega) \quad (7)$$

To reduce the sampling rate of the origin signal $x(n)$ by an integer factor M , the output sequence $y(n)$ is

$$y(n) = x(Mn) \quad (8)$$

The sampling period of $y(n)$ is increased to $\Delta t_y = M \cdot \Delta t_x$, and the DTLCT of $y(n)$ is

$$\tilde{Y}_{(a,b,c,d)}(\omega) = \frac{1}{M} \sum_{k=0}^{M-1} \tilde{X}_{(a,b,c,d)}\left(\frac{\omega - \text{sgn}(b)2\pi k}{M}\right) e^{j2\pi k b d \frac{\text{sgn}(b)\omega - \pi k}{(M\Delta t_x)^2}} \quad (9)$$

3. Identical relations

3.1. Identical relation in interpolation

In practical signal processing, we usually increase the sampling rate of the origin signal, and then remove its images by a filter. Due to the numerous operating works in LCT and we have to remove the images in LCT domain, we need to find a way to reduce the operations. Our target is the output signal $y(n)$, increased the sampling rate by L . Putting the LCT operation before the interpolation is a good way to reduce the operation. The low sampling rate in the origin signal can cause significantly decrease in LCT operation. Therefore, we have to find theorems to ensure that we can carry through this process with minus errors.

In this subsection, we will prove that the output signal interpolates before it goes through the filter $\tilde{H}_{(a,b,c,d)}(L\omega)$, will be the same as the one that interpolates after it goes through the filter $\tilde{H}_{(a,b,c,d)}(\omega)$, that is, $\tilde{Y}'_{(a,b,c,d)}(\omega) = \tilde{Y}_{(a,b,c,d)}(\omega)$. Equivalent in LCT domain indicates that the signal in time domain will

most likely be the same.

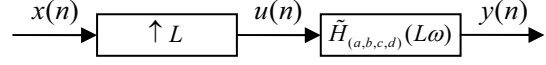


Fig.3.a

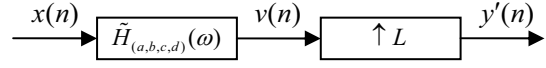


Fig.3.b

In Fig.3.a, according to the equation (7), we have

$$\tilde{U}_{(a,b,c,d)}(\omega_y) = \tilde{X}_{(a,b,c,d)}(L \cdot \omega_y) \quad (10)$$

Based on the convolution theorems of LCT [4], we have

$$\begin{aligned} \tilde{Y}_{(a,b,c,d)}(\omega) &= e^{-j\frac{d}{2b}\left(\frac{L\omega}{\Delta t_x}\right)^2} \tilde{U}_{(a,b,c,d)}(\omega) \cdot \tilde{H}_{(a,b,c,d)}(L \cdot \omega) \\ &= e^{-j\frac{d}{2b}\left(\frac{L\omega}{\Delta t_x}\right)^2} \tilde{X}_{(a,b,c,d)}(L \cdot \omega) \cdot \tilde{H}_{(a,b,c,d)}(L \cdot \omega) \end{aligned} \quad (11)$$

In Fig.3.b, based on convolution theorems of LCT, we have

$$\tilde{V}_{(a,b,c,d)}(\omega) = e^{-j\frac{d}{2b}\left(\frac{L\omega}{\Delta t_x}\right)^2} \tilde{X}_{(a,b,c,d)}(\omega) \cdot \tilde{H}_{(a,b,c,d)}(\omega) \quad (12)$$

Again according to the equation (7), we have

$$\tilde{Y}'_{(a,b,c,d)}(\omega) = \tilde{V}_{(a,b,c,d)}(L \cdot \omega) \quad (13)$$

Substitute (12) into (13), we obtain

$$\tilde{Y}'_{(a,b,c,d)}(\omega) = e^{-j\frac{d}{2b}\left(\frac{L\omega}{\Delta t_x}\right)^2} \tilde{X}_{(a,b,c,d)}(L \cdot \omega) \cdot \tilde{H}_{(a,b,c,d)}(L \cdot \omega) \quad (14)$$

Compare (11) with (14), we finally have

$$\tilde{Y}'_{(a,b,c,d)}(\omega) = \tilde{Y}_{(a,b,c,d)}(\omega) \quad (15)$$

Hence, the identical relation in interpolation can be derived as Fig.4

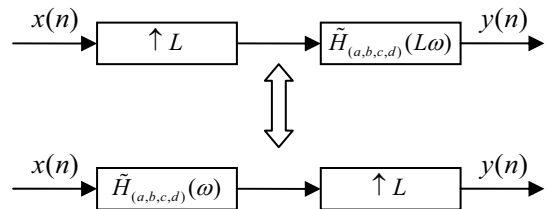


Fig. 4 Identical relation in interpolation

3.2. Identical relation in decimation

Like the method we deal with numerous operating works in interpolation, we have to find ways to reduce the operating works in decimation. Parallel to interpolation, we put the LCT operation in the low sampling rate side. In this subsection, we

will prove that the output signal decimates after it goes through the anti-aliasing filter $\tilde{H}_{(a,b,c,d)}(M\omega)$, will be the same as the one that decimates after it goes through the filter $\tilde{H}_{(a,b,c,d)}(\omega)$, that is, $\tilde{Y}'_{(a,b,c,d)}(\omega) = \tilde{Y}_{(a,b,c,d)}(\omega)$. Again, equivalent in LCT domain indicates that the signal in time domain will most likely be the same.

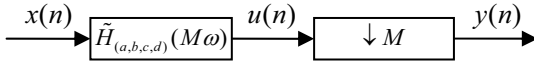


Fig.5.a

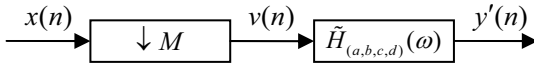


Fig.5.b

In Fig.5.a, based on the convolution theorems of LCT, we have

$$\tilde{U}_{(a,b,c,d)}(\omega) = e^{-j\frac{d}{2b}\frac{\omega^2}{(M\Delta t_x)^2}} \cdot \tilde{X}_{(a,b,c,d)}(\omega) \cdot H_{(a,b,c,d)}(M\omega) \quad (16)$$

According to equation (9), we have

$$\begin{aligned} \tilde{Y}_{(a,b,c,d)}(\omega) &= \frac{1}{M} \sum_{k=0}^{M-1} e^{j2\pi k d \frac{\text{sgn}(b)\omega - k\pi b}{(M\Delta t_x)^2}} \tilde{U}_{(a,b,c,d)}\left(\frac{\omega - \text{sgn}(b)2\pi k}{M}\right) \\ &= \frac{1}{M} \sum_{k=0}^{M-1} e^{j2\pi k d \frac{\text{sgn}(b)\omega - k\pi b}{(M\Delta t_x)^2}} \tilde{X}_{(a,b,c,d)}\left(\frac{\omega - \text{sgn}(b)2\pi k}{M}\right) \cdot \tilde{H}_{(a,b,c,d)}(\omega - \text{sgn}(b) \cdot 2\pi k) \cdot e^{-j\frac{d}{2b}\frac{(\omega - \text{sgn}(b)2\pi k)^2}{(M\Delta t_x)^2}} \end{aligned} \quad (17)$$

Because the filter function $\tilde{H}_{(a,b,c,d)}(\omega)$ has the following property:

$$\begin{aligned} \tilde{H}_{(a,b,c,d)}(\omega) &= \tilde{H}_{(a,b,c,d)}(\omega - \text{sgn}(b) \cdot 2\pi b) \cdot e^{j2\pi d \frac{\text{sgn}(b)\omega - \pi b}{(M\Delta t_x)^2}} \\ &= \tilde{H}_{(a,b,c,d)}(\omega - k \cdot \text{sgn}(b) \cdot 2\pi b) \cdot e^{j2\pi k d \frac{\text{sgn}(b)\omega - k\pi b}{(M\Delta t_x)^2}} \end{aligned} \quad (18)$$

Using (18), the equation (17) can be reduced to

$$\tilde{Y}_{(a,b,c,d)}(\omega) = \frac{1}{M} \sum_{k=0}^{M-1} \tilde{X}_{(a,b,c,d)}\left(\frac{\omega - \text{sgn}(b)2\pi k}{M}\right) \cdot \tilde{H}_{(a,b,c,d)}(\omega) \cdot e^{-j\frac{d}{2b}\frac{(\omega - \text{sgn}(b)2\pi k)^2}{(M\Delta t_x)^2}} \quad (19)$$

In Fig.3.b, according to the equation (9), we obtain

$$\tilde{V}_{(a,b,c,d)}(\omega) = \frac{1}{M} \sum_{k=0}^{M-1} e^{j2\pi k d \frac{\text{sgn}(b)\omega - k\pi b}{(M\Delta t_x)^2}} \tilde{X}_{(a,b,c,d)}\left(\frac{\omega - \text{sgn}(b)2\pi k}{M}\right) \quad (20)$$

Based on the convolution theorems of LCT, we have

$$\begin{aligned} \tilde{Y}'_{(a,b,c,d)}(\omega) &= \tilde{V}_{(a,b,c,d)}(\omega) \tilde{H}_{(a,b,c,d)}(\omega) \cdot e^{-j\frac{d}{2b}\frac{\omega^2}{(M\Delta t_x)^2}} \\ &= \frac{1}{M} \sum_{k=0}^{M-1} e^{j2\pi k d \frac{\text{sgn}(b)\omega - k\pi b}{(M\Delta t_x)^2}} \tilde{X}_{(a,b,c,d)}\left(\frac{\omega - \text{sgn}(b)2\pi k}{M}\right) \cdot \tilde{H}_{(a,b,c,d)}(\omega) \cdot e^{-j\frac{d}{2b}\frac{\omega^2}{(M\Delta t_x)^2}} \\ &= \frac{1}{M} \sum_{k=0}^{M-1} e^{-j\frac{d}{2b}\frac{\omega^2}{(M\Delta t_x)^2} + j\frac{d}{2b}\frac{4\pi k \text{sgn}(b)\omega - 4\pi^2 k^2 b^2}{(M\Delta t_x)^2}} \tilde{X}_{(a,b,c,d)}\left(\frac{\omega - \text{sgn}(b)2\pi k}{M}\right) \cdot \tilde{H}_{(a,b,c,d)}(\omega) \\ &= \frac{1}{M} \sum_{k=0}^{M-1} e^{-j\frac{d}{2b}\frac{(\omega - \text{sgn}(b)2\pi k)^2}{(M\Delta t_x)^2}} \tilde{X}_{(a,b,c,d)}\left(\frac{\omega - \text{sgn}(b)2\pi k}{M}\right) \cdot \tilde{H}_{(a,b,c,d)}(\omega) \end{aligned} \quad (21)$$

Compare (19) with (21), we can find that

$$\tilde{Y}'_{(a,b,c,d)}(\omega) = \tilde{Y}_{(a,b,c,d)}(\omega) \quad (22)$$

Hence, we have proved the identical relation in decimation like Fig.6

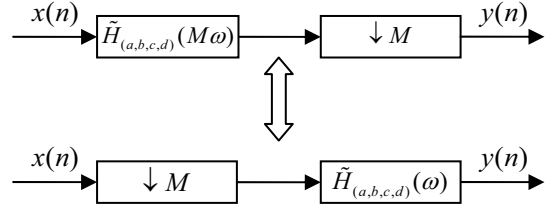


Fig.6 Identical relation in decimation

3.3. Special cases

When the parameters of the linear canonical transform reduce to $(a, b, c, d) = (\cos \theta, \sin \theta, -\sin \theta, \cos \theta)$, the above mentioned identical relation for the decimation and interpolation reduces to the results of fractional Fourier transform domain [2], which, with the parameters $(a, b, c, d) = (0, 1, -1, 0)$, becomes the classical result in Fourier domain [1].

It is shown in this sense that the classical results in Fourier transform domain and the fractional Fourier transform domain are all special cases of our results.

4. Simulation results

In order to show the correctness of the derived results, the simulation results of the proposed theorems are presented in this section.

4.1. Simulation results of interpolation

The input signal and its LCT are plotted in the following figure 7.

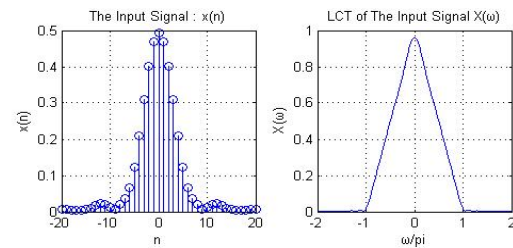


Fig.7 Input Signal and its LCT for interpolation

In simulation, the parameter of the LCT is chosen as $(a, b, c, d) = (1, 2/\pi, 0, 1)$, the sampling time is $[-0.1s, 0.1s]$, and the sampling frequency is 200Hz . The interpolation factor is $L = 4$, after the interpolation of the above mentioned signal, the output signal and its LCT is plotted in Fig.8

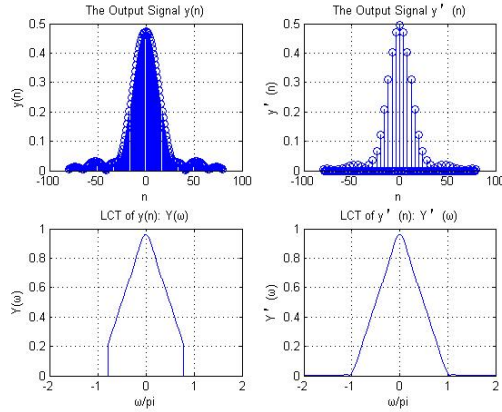


Fig.8 Output signal and its LCT after interpolation by $L = 4$

In Fig.8, $y(n)$ and $y'(n)$ represent the corresponding output signal in Fig.3.a and b, respectively. Where $y(n)$ is the output signal that interpolates before passing through the filter, while $y'(n)$ is the output signal that interpolates after passing through the filter. $Y(\omega)$ and $Y'(\omega)$ are their corresponding LCT in LCT domain. It can be concluded from the simulation results that the identical relation in interpolation is viable.

4.2. Simulation results of decimation

In simulation of decimation, we have to create input signal with higher sampling frequency and more sampling points. We assume the sampling time is $[-0.1s, 0.1s]$, the sampling frequency is $1kHz$, and the LCT parameter is $(a, b, c, d) = (1, 2/\pi, 0, 1)$. The input signal and its LCT are showed in Fig.9.

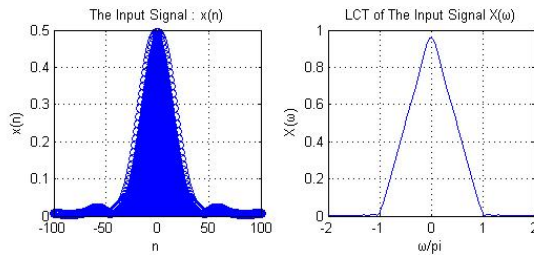


Fig.9 Input Signal and its LCT for decimation

After decimation by an integer $M = 3$, we can get the output signal and its LCT in Fig.10. $y(n)$ and $y'(n)$ represent the corresponding output signal in Fig.5.a and b, respectively. Where $y(n)$ is the output signal that decimates after passing through the filter $\tilde{H}_{(a,b,c,d)}(M\omega)$, while $y'(n)$ is the output signal that decimates before passing through the filter $\tilde{H}_{(a,b,c,d)}(\omega)$. $Y(\omega)$ and $Y'(\omega)$ are their corresponding LCT in the LCT domain. The simulation results reveal that these two LCT are the same. As a result, the identical relation in decimation is proved to be valid.

5. Conclusion

In this paper, two identical relations in sampling rate conversion are present and their validity is proved both in theory and simulations. These two relations can be widely used in

practical application for its high efficiency and low requirement for system capacity.

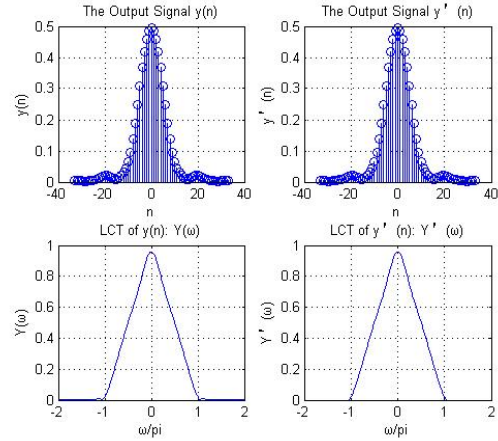


Fig.10. Output signal and its LCT after decimation by $M = 3$

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